SVKM's NMIMS
Mukesh Patel School of Technology Management and Engineering
Electronics & Telecommunication Department

Program: B Tech (EXTC/Computer Science/ CSE(DS) 311 (VT)/B Tech & MBA Tech (Information Technology/Computer Engineering)/BTI (Computer Engineering/ Information Technology)				**Semester: V / VI / VIII/X /XII**	
Course : Image and Video Processing				**Code:**	
Teaching Scheme				**Evaluation Scheme**	
Lecture (Hours per week)	**Practical (Hours per week)**	**Tutorial (Hours per week)**	**Credit**	**Internal Continuous Assessment (ICA) (Marks - 50)**	**Term End Examinations (TEE) (Marks - 100)**
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50

Pre-requisite: Basic operations on 2D arrays

Course Objective

This course introduces concepts, methodologies and performance metrics for still image and motion picture processing. It also helps to develop a foundation for further study and research in the signal processing domain.

Course Outcomes

After completion of the course, students will be able to -

1. **Explain** the fundamentals of digital images, including pixel relationships, resolution, distance measures, and file formats. *(L2 - Understanding)*
2. **Apply** image enhancement techniques such as point processing, spatial filtering, and histogram equalization for image improvement. *(L3 - Applying)*
3. **Analyze** morphological image processing techniques and edge detection methods for feature extraction. *(L4 - Analyzing)*
4. **Compare** different image segmentation methods, including thresholding, region-based segmentation, and the Otsu method. *(L4 - Analyzing)*
5. **Explain** video representation concepts and motion estimation techniques used in digital video compression. *(L2 - Understanding)*

Detailed Syllabus

Unit	Description	Duration
1.	**Image fundamentals** Representation of digital image, image as a matrix, image as a function, basics of colour image, spatial and gray level resolution, distance measures-Euclidean distance and m path, image file formats (TIFF, PNG, JPEG).	**02**
2.	**Image enhancement** **Point processing techniques** Linear- digital negative, Piecewise linear-contrast stretching, thresholding, intensity level slicing, Nonlinear- log transformation, power law transformation, **Neighborhood processing techniques**-smoothing spatial filters, sharpening spatial filters, median filters, Gaussian filtering. **Histogram processing**-histogram equalization. **Image in Frequency domain**: Fourier spectrum, Fourier analysis, filtering in frequency domain	**06**
3.	**Image transforms** Discrete Fourier Transform, Hadamard transform, Discrete cosine transform and its applications for image compression	**02**

4.	**Morphological image processing** Dilation, erosion, opening, closing, Hit–or-Miss transformation, basic morphological algorithms- boundary extraction on binary images, thinning, thickening, gray scale morphology	**04**
5.	**Edge Detection** Gradient based edge detection- First order and second order derivative, Edge detection operators- Prewitt, Sobel, Laplacian, Laplacian of Gaussian, canny edge detector.	**02**
6.	**Image segmentation** Basic relationships between pixels, Component labeling (4, 8 and m connectivity) Segmentation using thresholding, Bayesian classification of foreground and background pixels, expectation- maximization theorem for binarization, Otsu method, region-based segmentation- region growing, region splitting and merging.	**04**
7	**Colour Image Processing** Colour fundamentals and colour models (RGB, CMY, CMYK, HSI, YUV, HSV), model transformation, colour slicing, histogram processing of colour images.	**04**
8	**Fundamentals of digital video and its compression** Video representation- temporal correlation, video frame classifications, I, P and B frames, Fundamentals of motion estimation, motion estimation algorithms- exhaustive search block matching, 2D-log search method and 3 step search method.	**06**
	Total	**30**

Text Books

1. R.C Gonzalez and Richard Woods, *Digital Image Processing*, Pearson publication, 4th Edition, 2018.
2. Wilhelm Burger and Mark J. Burge, *Digital Image Processing: An Algorithmic Introduction*, Springer publications, 2nd Edition, 2022.

Reference Books

1. Alasdair McAndrew, *A Computational Introduction to Digital Image Processing*, CRC Press, 2nd Edition, 2021
2. Syed Jahangir Badashah, *Digital Image Processing*, Lambert Academic Publishing, 2020

Laboratory Work

8 to 10 practical exercises (and a practicum) based on the syllabus.

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Signature

(Head of the Department)